



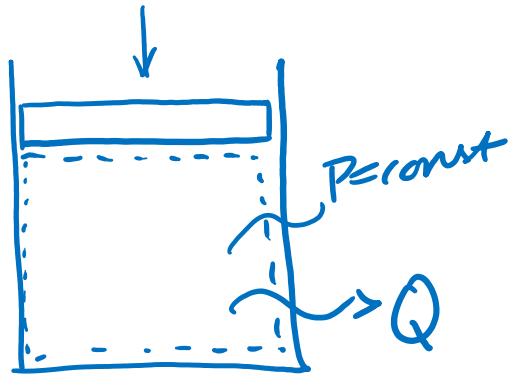
ME 300:
Engineering Thermodynamics

Introduction to Phase Change

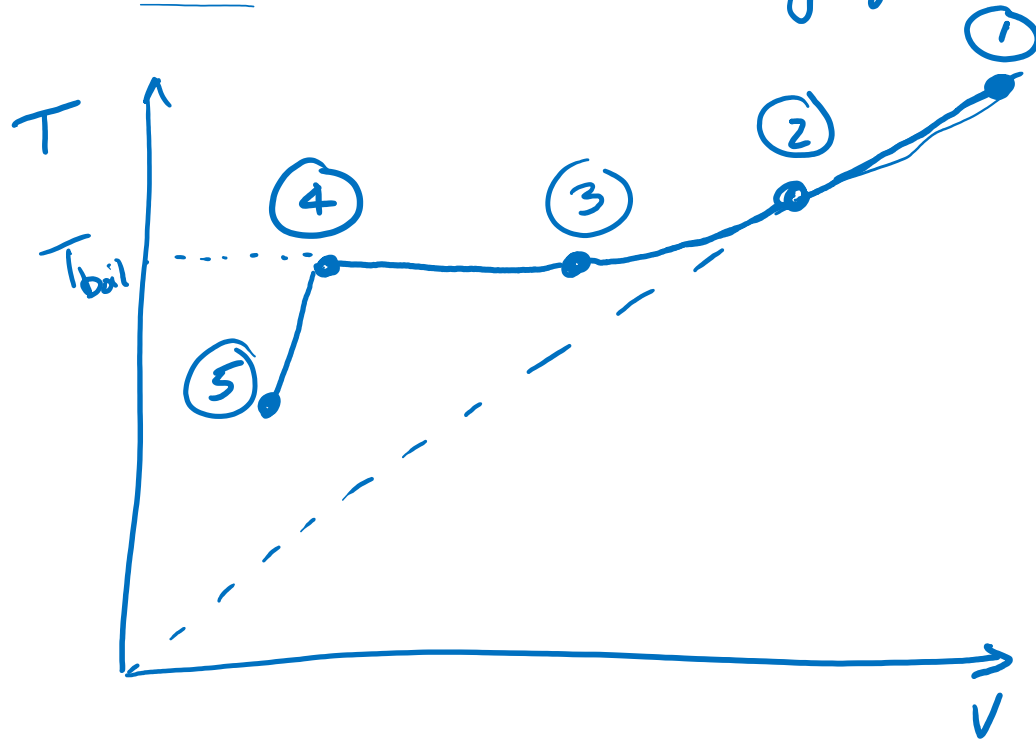
Why do we get phase changes?

→ ideal gases → $u_{\text{mol}} \ll u_{\text{b/w mol}}$

→ Intermolecular forces were negligible



ideal: $Pv = \underline{RT}$



1-2: ideal gas

2-3: non-ideal

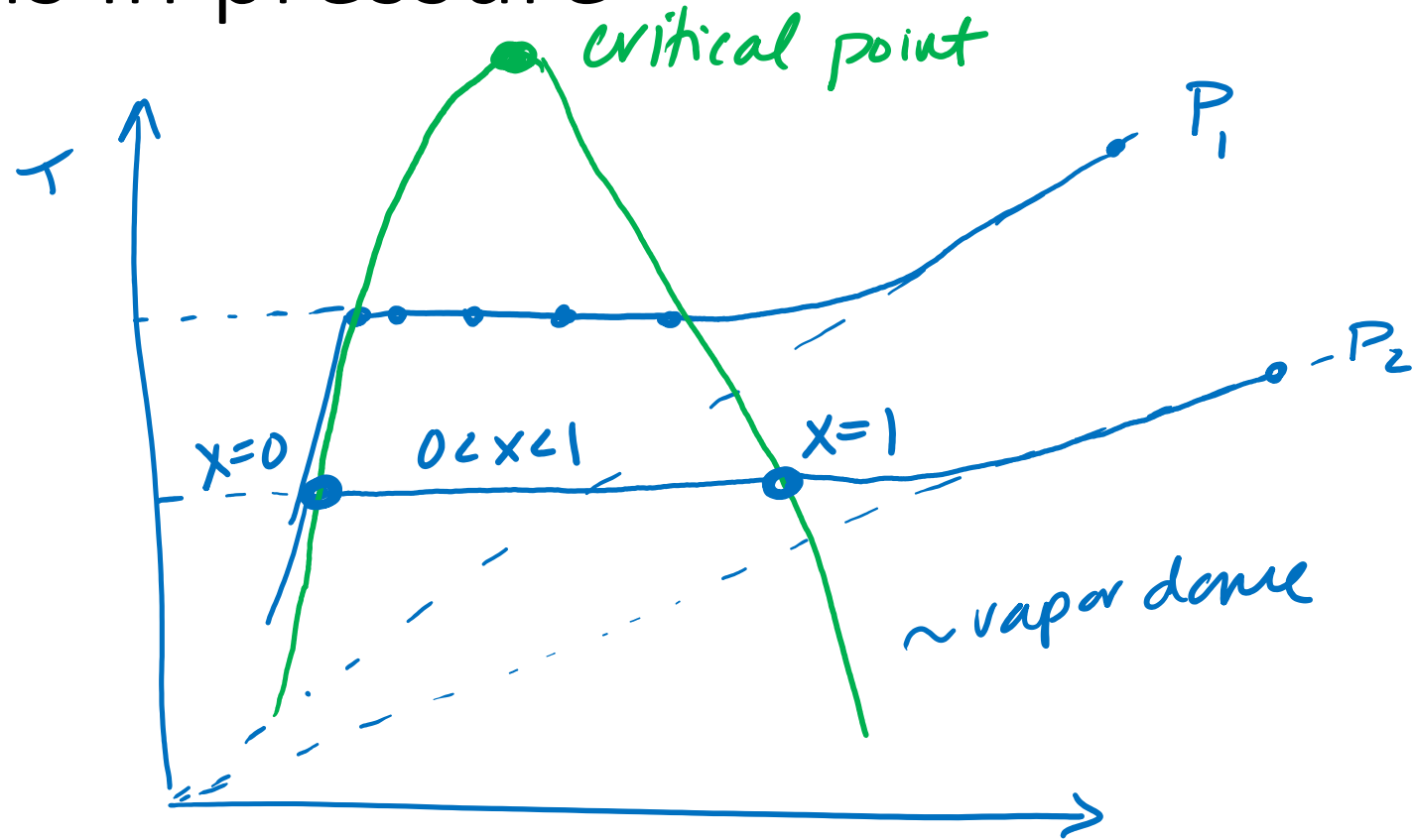
3 ⇒ form liquid

3-4: first u_f to
all liq.

4 ⇒ all liquid

5 ⇒ cooled liquid.

Variations in pressure



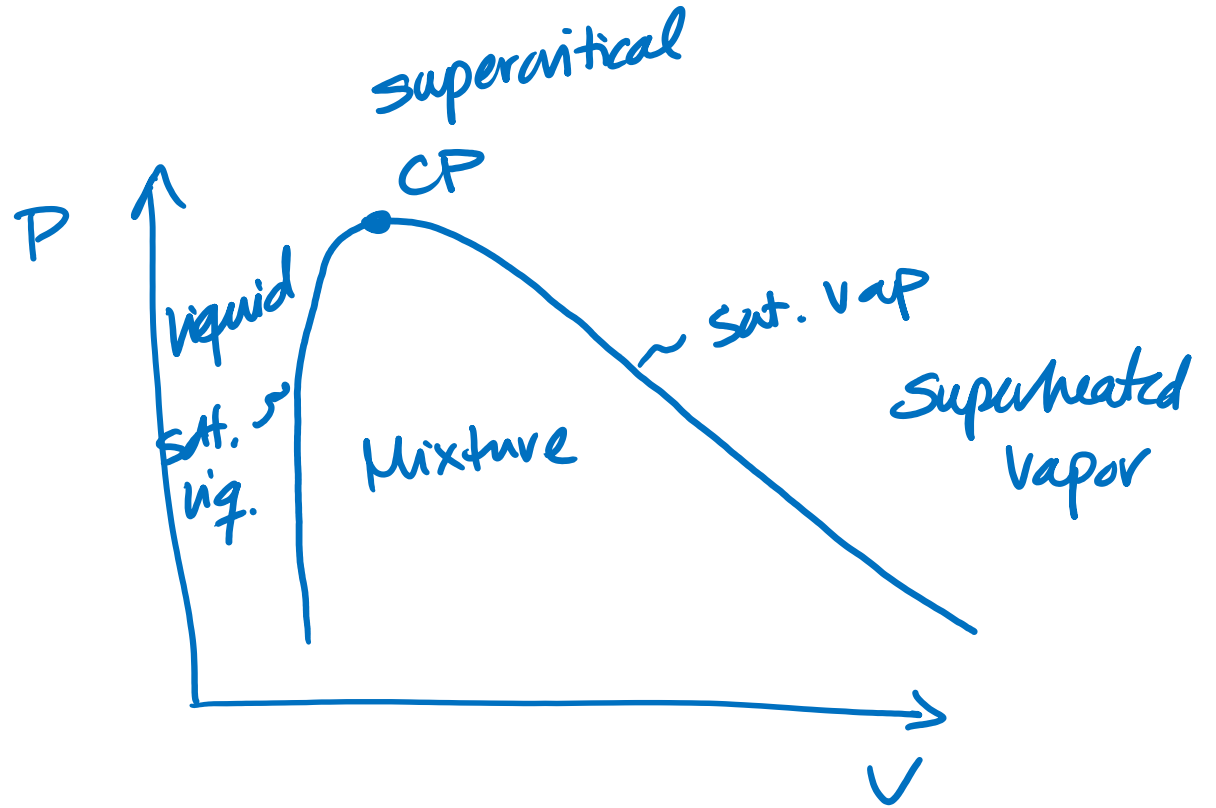
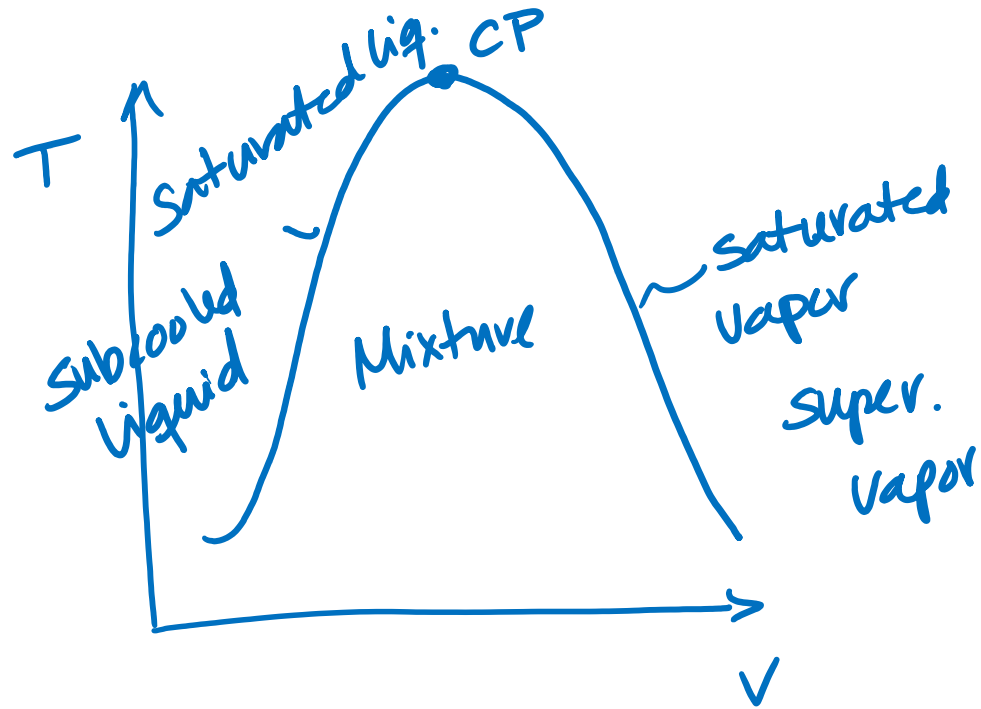
★ inside dome
(mixture of vap. &
liquid) $\Rightarrow$

P and T are
not independent

Inside dome $\rightarrow$ Quality $x = \frac{M_{\text{vapor}}}{M_{\text{vapor}} + M_{\text{liquid}}} = \frac{M_g}{M_g + M_f}$

Quality

T-v and P-v diagrams



$$x = \frac{M_g}{M_g + M_f}$$